

Objective:

The main objective of my project is to build a system that can predict how severe a disease outbreak might become, whether mild, moderate, or severe. By using machine learning and deep learning models on real outbreak data from Toronto healthcare institutions, I aim to give public health teams a tool that helps them prepare faster, respond better, and manage resources more effectively during outbreaks.

Background:

For my Major Research Project, I focused on predicting the severity of disease outbreaks using computational methods to support public health systems. Outbreaks like COVID-19 showed how quickly healthcare systems can become overwhelmed, especially when time is limited. Traditional approaches often failed to capture real-time patterns, spatial spread, or the interactions of different risk factors. To address this, I explored machine learning and deep learning models.

In the literature review, I examined how researchers have applied different approaches which included LSTM, CNN, BiLSTM, Random Forest, XGBoost, and also hybrid models like CNN-LSTM for forecasting outbreak severity. I also studied how these models handled challenges such as missing data, feature selection, model interpretability, and generalizability.

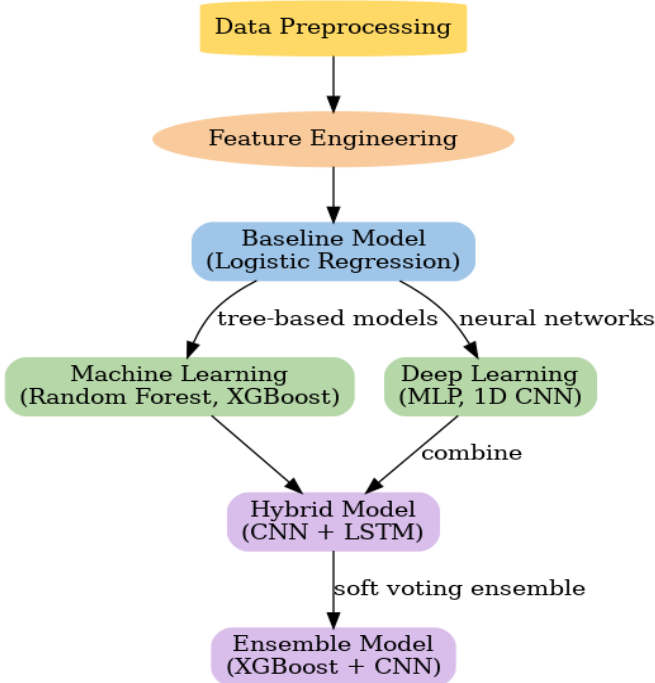
The goal of this review was to identify which techniques were most effective and where research still have gaps. These perceptions guided my own project, in which I developed a predictive model for assessment of outbreak severity using real-world data that came from healthcare institutions located in Toronto.

Methodology:

To begin, I set up a baseline model using Logistic Regression. This simple and fast method helped me establish a reference point for comparing more advanced models. I divided the data into training validation and test sets. After that , I applied standard preprocessing such as scaling and imputing missing values and then trained the model.

Next, I moved on to machine learning and deep learning models. For ML, I focused on Random Forest and XGBoost since they strongly perform on structured healthcare data. XGBoost captures complex non-linear patterns as a powerful increasing method beyond others, while Random Forest reduces overfitting by combining many decision trees. For Deep Learning, I used a Multilayer Perceptron (MLP) as well as a 1D Convolutional Neural Network (CNN). CNN was selected in order to capture local feature interactions that customary models might happen to miss, while the MLP works quite well with tabular data.

Finally, I experimented with hybrid and ensemble approaches. I built a CNN-LSTM hybrid model to combine spatial and temporal pattern learning, though its performance was limited without further tuning. To improve stability, I also created an ensemble of XGBoost and CNN using soft voting, which balanced their strengths. This ensemble gave more consistent results and showed that combining models can reduce weaknesses of individual approaches.



Results:

I began with Logistic Regression as a baseline, which gave surprisingly strong results at 98% accuracy with high precision and recall. This showed that even a simple linear model could capture useful patterns in the data.

Random Forest improved on this with 99.6% accuracy and excellent balance across metrics, proving to be both reliable and consistent. XGBoost then went a step further, achieving perfect scores (100%) across accuracy, precision, recall, and F1. This confirmed its strength in handling structured data and complex feature interactions.

On the other hand, deep learning models simply faltered. The Multilayer Perceptron (MLP) only reached up to about 72% accuracy, while the CNN was at even lower levels near 70%. The CNN+LSTM hybrid performed worst, at under 70% accuracy. These models displayed diminished precision and recall because the dataset lacked sequential or spatial patterns they commonly leverage.

Finally, I made a soft-voting ensemble model of CNN along with XGBoost. Although it didn’t exceed XGBoost, it achieved ~99.7% accuracy upon both validation and test sets, which showed stable and well-generalized performance.

	Logistic Regression				
Evaluation Metrics	(Baseline)	Random Forest	XGBoost	MLPClassifier	CNN
Accuracy	0.9803	0.9956	1.0000	0.7158	0.7071
Macro Precision	0.9895	0.9979	1.0000	0.4520	0.5555
Macro Recall	0.9508	0.9892	1.0000	0.5784	0.3886
Macro F1-score	0.9690	0.9935	1.0000	0.4897	0.3846
AUC-ROC (macro)	0.9990	1.0000	1.0000	0.8636	0.7006
AUC-PR (macro)	0.9969	1.0000	1.0000	0.7266	0.4905
Log Loss	0.1128	0.0895	0.0072	0.8046	0.7332
Cohen's Kappa	0.9564	0.9905	1.0000	0.4205	0.1313
Matthews CorrCoef	0.9571	0.9905	1.0000	0.4427	0.1786

	CNN + LSTM	XGB + CNN	XGB + CNN
Evaluation Metrics	(Hybrid)	(Ensemble Val)	(Ensemble Test)
Accuracy	0.6962	0.9978	0.9967
Macro Precision	0.3365	0.9990	0.9984
Macro Recall	0.3370	0.9946	0.9928
Macro F1-score	0.2873	0.9968	0.9956
AUC-ROC (macro)	0.6923	1.0000	1.0000
AUC-PR (macro)	0.4698	1.0000	1.0000
Log Loss	0.7439	0.2536	0.2542
Cohen's Kappa	0.0125	0.9952	0.9929
Matthews CorrCoef	0.0337	0.9953	0.9929

Conclusions:

From this project, I learned that tree-based models worked the best for predicting outbreak severity. XGBoost stood out with almost perfect accuracy, and Random Forest was also very reliable. My starting point, Logistic Regression, did fairly well but couldn’t capture the more complex patterns. On the other hand, deep learning models like MLP, CNN, and CNN+LSTM didn’t perform as well, likely because the dataset was small and didn’t have the kinds of patterns these models usually need.

When I combined XGBoost with CNN into an ensemble model, the results were even more stable and close to perfect. This showed that blending models can make predictions stronger and more balanced.

In the future, I’d like to expand the dataset with additional sources such as environmental as well as demographic data, so public health teams can trust and use the predictions in real time.